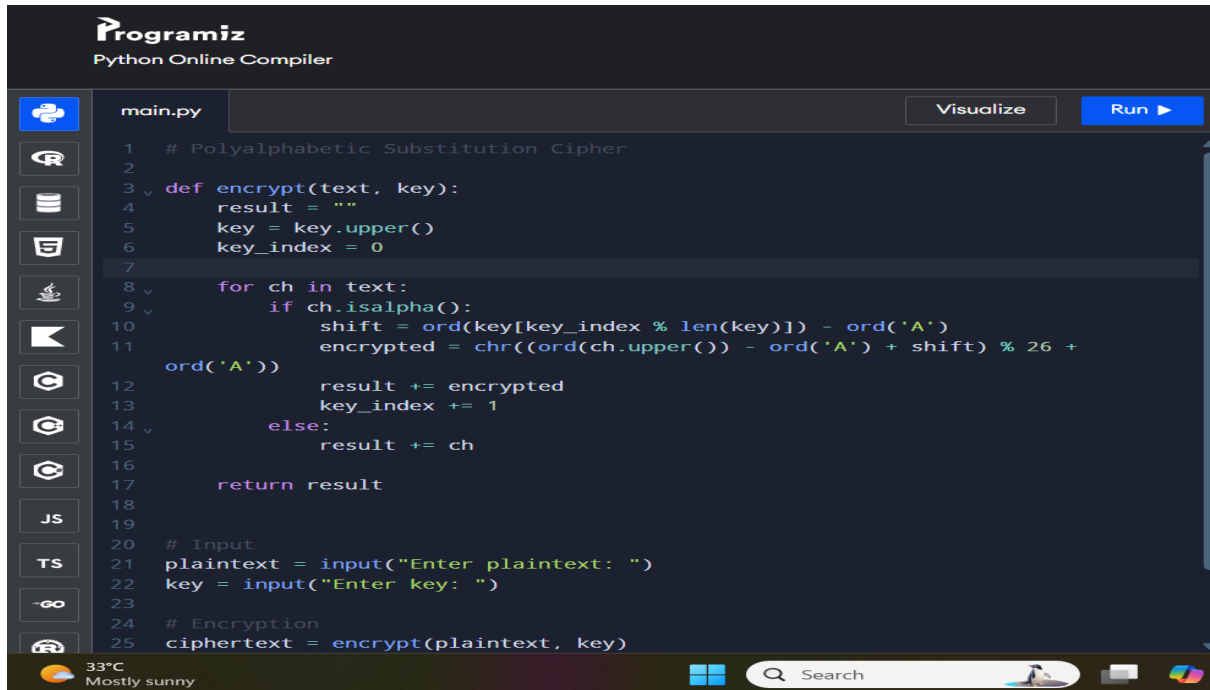


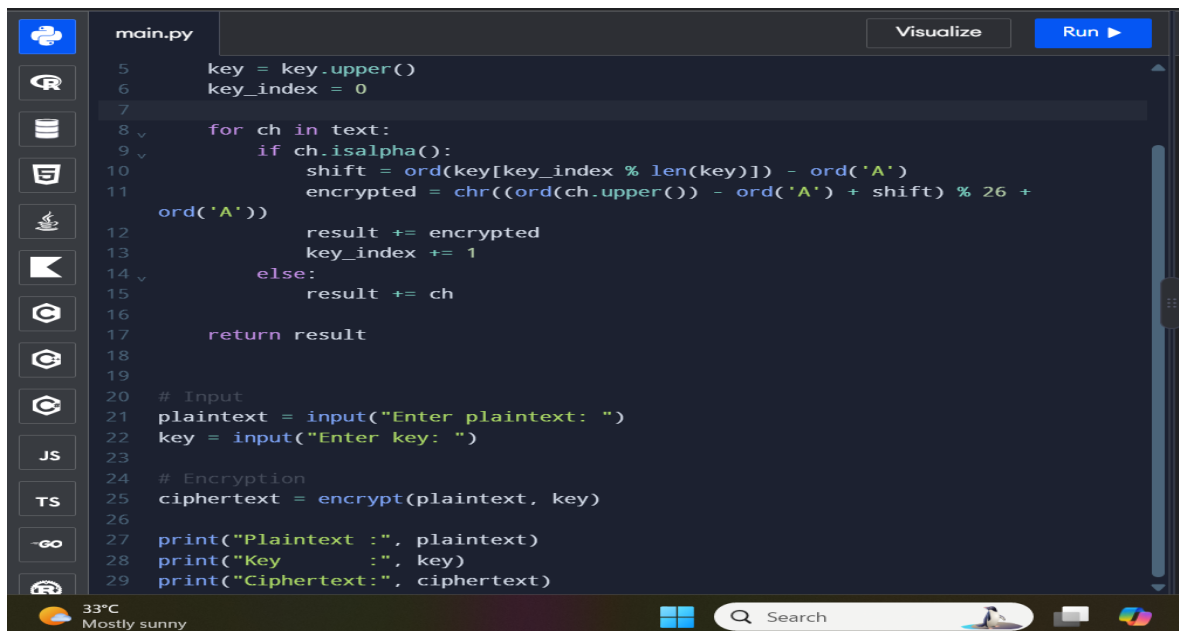
EXPERIMENT-4

4. Write a C program for polyalphabetic substitution cipher uses a separate monoalphabetic substitution cipher for each successive letter of plaintext, depending on a key.



The screenshot shows the Programiz Python Online Compiler interface. The file is named 'main.py'. The code defines an 'encrypt' function that takes 'text' and 'key' as arguments. It converts the key to uppercase and initializes a 'key_index' to 0. A loop iterates over each character in the text. If the character is an alphabet, it calculates a shift based on the current key character and the character's position in the alphabet, then shifts the character accordingly. Non-alphabet characters are added to the result as-is. The function returns the encrypted result. Below the function, there is input for plaintext and key, and a call to the 'encrypt' function to produce the ciphertext.

```
1 # Polyalphabetic Substitution Cipher
2
3 def encrypt(text, key):
4     result = ""
5     key = key.upper()
6     key_index = 0
7
8     for ch in text:
9         if ch.isalpha():
10            shift = ord(key[key_index % len(key)]) - ord('A')
11            encrypted = chr((ord(ch.upper()) - ord('A') + shift) % 26 +
12                           ord('A'))
13            result += encrypted
14            key_index += 1
15        else:
16            result += ch
17
18    return result
19
20 # Input
21 plaintext = input("Enter plaintext: ")
22 key = input("Enter key: ")
23
24 # Encryption
25 ciphertext = encrypt(plaintext, key)
```



This screenshot shows the same code as the previous image, but with additional print statements at the end to display the results. The code now prints the plaintext, the key, and the resulting ciphertext.

```
5     key = key.upper()
6     key_index = 0
7
8     for ch in text:
9         if ch.isalpha():
10            shift = ord(key[key_index % len(key)]) - ord('A')
11            encrypted = chr((ord(ch.upper()) - ord('A') + shift) % 26 +
12                           ord('A'))
13            result += encrypted
14            key_index += 1
15        else:
16            result += ch
17
18    return result
19
20 # Input
21 plaintext = input("Enter plaintext: ")
22 key = input("Enter key: ")
23
24 # Encryption
25 ciphertext = encrypt(plaintext, key)
26
27 print("Plaintext :", plaintext)
28 print("Key       :", key)
29 print("Ciphertext:", ciphertext)
```

